



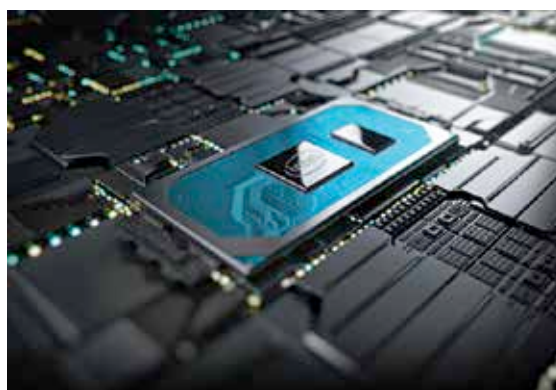
Mithun Mohandas |
mithun@digit.in

Buying smart doesn't necessarily mean that you should get the cheapest possible device. Spending a little extra and buying something that's bound to last you for a long time is more prudent than buying a new device every other year. There are also ways to maintain a device better so that you don't end up losing out on performance. In last month's cover story, we explored the numerous ways in which you could save money while shopping while also learning how to take care of the gadgets you already own. This month, we're going to tell you what to keep in mind while deciding on which products to buy. We'll look at the features that matter the most for improving your experience and productivity, and the ones that can be treated as luxury. We'll start off with laptops.

1. PROCESSOR

Often, you won't see much of the processors' specifications being listed

on the product page when shopping online and the same can be said for when you're checking out products at the store. What is mentioned is usually the frequency or the speed of the processor. While the processor frequency is an important factor, it should never be looked at as a single factor. The processor frequency along



Frequency and cores are just part of the picture

with the IPC (Instructions Per Clock) is what contributes to the overall performance. IPC figures are not publicly available in an easy to consume format so the best course of action is to check out comparative reviews of the processor to see where it sits.

There are situations where an older processor with higher core counts are beaten by newer processors with fewer cores. The reason is usually the architectural improvements in the way the processor is designed. Also, the operating system plays a huge role in how the workload is assigned to different cores on your processor. For example, the newer Intel 12th Gen processors perform better when paired with Windows 11 because of how the process scheduler in Windows 11 works. The difference isn't much, but it does matter. When looking for a laptop, try to opt for a model which sports a newer processor and check out review websites such as [digit.in](https://www.digit.in) to see which processor is better and by what margin. Don't simply go by speed or the number of cores.

2. RAM

Quite a few laptops ship with 8 GB of RAM with the option of expanding the memory pool by adding an additional memory module later on. Because of this, laptops often ship with just one memory module. This can be verified by running simple software such as CPU-Z which will let you know the number of modules installed. If you've only got one module installed, then the memory is operating in single-channel mode.

This can have a negative impact on the performance of the system by a magnitude of up to 40 per cent. This is with respect to laptops with DDR4 memory. DDR5 has two channels within each module and the performance impact isn't as drastic.

Another aspect that needs to be checked is the Rank of the memory. Rank is basically the way memory is organised

on a logical level. When there are more Ranks within a module, the memory controller – which decides where data is to be stored and how it can be retrieved – gets stressed and there is a minor performance penalty.